

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
2nd Semester of B. Pharm. Examination
University Theory Examination April- May 2018
BPH2002 Pharmaceutical Organic Chemistry-I

Date: 09.05.18, Wednesday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 75

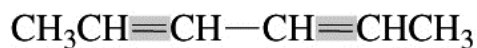
Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

Q 1 Attempt all questions. Each question is of one mark. 20

1. All statement is correct for SN₂ reaction. Except one
[A] Follow Second order kinetic
[B] Rearrangement is possible
[C] Inversion of configuration takes place
[D] Single step reaction
2. Reaction of Secondary Alkyl Halide with good nucleophile and strong base _____ mechanism is follow.
[A] E₁
[B] SN₁
[C] E₂
[D] SN₂
3. Addition of 2Br₂ to Acetylene will give _____.
[A] 1,1,2,2 - tetrabromo ethane
[B] 1,2 - dibromo ethene
[C] 1,2 - dibromo ethane
[D] 2,2- dibromo ethane
4. Identify the type of following diene.

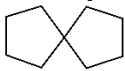


A



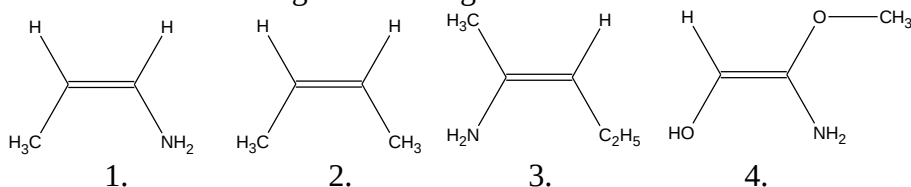
B

- [A] A is Isolated and B is Conjugated
[B] A is Conjugated and B is Isolated
[C] A is Cumulated and B is Isolated
[D] A is Conjugated and B is Cumulated
5. Aldehyde and ketone gives _____ type of reaction
[A] Substitutions
[B] Addition
[C] Elimination
[D] Rearrangement

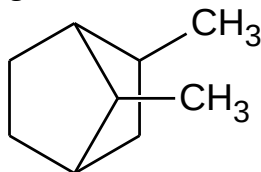
6. Hydrazine hydrate reagent is used in _____ reaction.
[A] Aldol Condensation
[B] Wolf-kishner reduction
[C] Clemmensen reduction
[D] Wittig reaction
7. Lucas test is used to differentiate _____.
[A] Alcohol
[B] Amine
[C] Phenol
[D] Anilide
8. Which of the following is not a primary amine?
[A] 3-Hexylamine
[B] 1-Butylamine
[C] Diethylamine
[D] Cyclohexylamine
9. Which pairing of general formula and compound class is incorrect?
[A] RNH_2 ; amine
[B] RCONH_2 ; amide
[C] R_3N ; tertiary amine
[D] RC=NH ; imide
10. Identify the following structure;

[A] Spiro
[B] Bicyclic
[C] Fused
[D] Bridgehead
11. Which reagent is a good nucleophile?
[A] Ammonia
[B] Water
[C] *tert*-Butoxide
[D] Hydroxy
12. Which of the following is most basic?
[A] Aniline
[B] Triethyl amine
[C] Ammonia
[D] Ethyl amine
13. Which of the following is NOT polar aprotic solvent?
[A] DMSO
[B] Acetonitrile
[C] Acetone
[D] Dihydrogen Monoxide
14. Which of the following has least p^{K_a} value and more acidic?
[A] Acetic acid
[B] Chloroacetic acid
[C] Fluoroacetic acid
[D] Difluoroacetic acid

15. 2,2-dimethylpropane is IUPAC name of _____
 [A] Neopentane
 [B] Isopentane
 [C] Isobutane
 [D] Isopropane
16. Which molecule contains a *SP*-hybridized atom?
 [A] Water
 [B] Ammonia
 [C] Acetylene
 [D] Ammonia
17. Which of the following CANNOT be made in good yield from $\text{CH}_3\text{CH}_2\text{Cl}$ using an $\text{S}_\text{N}2$ reaction?
 [A] $\text{CH}_3\text{CH}_2\text{OC}(\text{CH}_3)_3$
 [B] $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$
 [C] $\text{CH}_3\text{CH}_2\text{OH}$
 [D] $\text{CH}_3\text{CH}_2\text{SH}$

18. Which of the following has E configuration?



- [A] 1
 [B] 2
 [C] 3
 [D] 4
19. Identify the IUPAC of following structure;

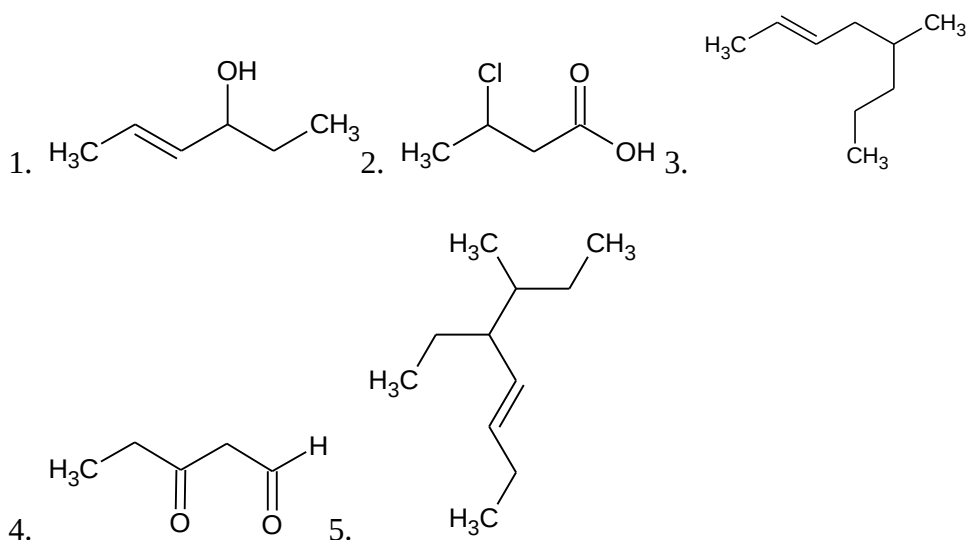


- [A] 2,7-dimethylbicyclo[2.2.1]heptane
 [B] 2,7-dimethylbicyclo[1.2.2]heptane
 [C] 2,7-dimethylbicyclo[2.2.1]nonane
 [D] 2,7-dimethylbicyclo[1.2.2]nonane
20. Identify the following structural isomer.
 $\text{CH}_3 - \text{CH}_2 - \text{O} - \text{CH}_2 - \text{CH}_3$
 Ethoxyethane
 $\text{CH}_3 - \text{O} - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$
 1-Methoxypropane

- [A] Chain Isomer
 [B] Tautomer's
 [C] Metamerism
 [D] Functional Isomer

SECTION – II

- Q 2** Explain reaction, mechanism, reactivity, kinetic and stereochemistry of SN_2 reaction. Write various factors affecting rate of SN_2 reaction. (10)
- Q 3** Attempt any TWO of the following (10)
- A** Write a note on structural isomerism in organic compounds. 05
- B** Explain reaction, mechanism, reactivity, kinetic and orientation of E_1 reaction. 05
- C** Give IUPAC name of following structure; 05



SECTION – III

- Q 4** Attempt any ONE of the following (10)
- A** Write any three preparation of alkane and alkene. Explain free radical mechanism for halogenation of alkane. 10
- B** Write reaction and mechanism for following reaction. 10
- (i) Aldol Condensation (ii) Cannizzaro Reaction
- Q 5** Attempt any FIVE of the following (25)
- A** Write any three preparation and reaction of alcohol. 05
- B** Explain hofmann degradation of amides with suitable reaction and mechanism. 05
- C** Write down any three methods for preparation of aldehydes and ketones. 05
- D** Write a note on malonic ester synthesis. 05
- E** Write Markovnikov's and Saytzeff's rules with suitable examples. 05
- F** Write structure and use of following: 05
1. Amphetamine 2. Chloral Hydrate 3. Chloroform 4. Ethyl Alcohol
5. Salicylic acid